

The image displays four Feynman diagrams representing the optical theorem for a two-point function. Each diagram shows a vertical dashed red line representing a cut. The diagrams are summed together and equated to an integral expression.

The first diagram shows a two-point function with external momenta p_1 and p_2 . The cut is vertical. The second diagram shows the same two-point function with external momenta $k-p_1$ and p_2 . The third diagram shows the same two-point function with external momenta p_1 and $k-p_2$. The fourth diagram shows the same two-point function with external momenta $k-p_1$ and $k-p_2$.

The sum of these diagrams is equal to the following integral expression:

$$= \text{disc}_{p_1^2} \text{disc}_{p_2^2} \int \frac{d^d k}{(2\pi)^d} \frac{N_{\boxtimes}^{q\bar{q}} \delta_{q^2}^+(p_{12} - k) \delta_0^+(k)}{p_1^2 p_2^2 (k - p_2)^2 (k - p_1)^2}$$